

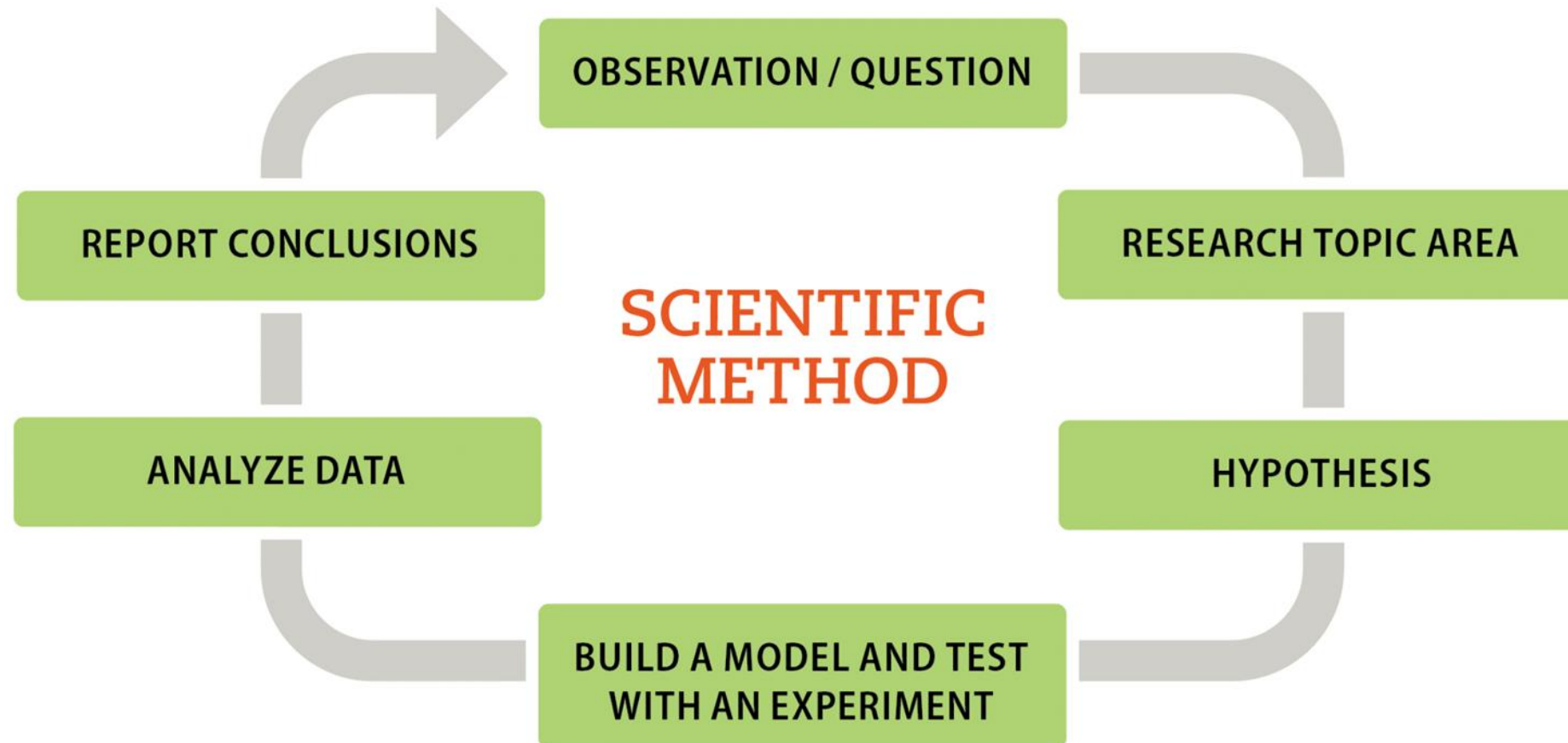
Dissertation Preparation

Context, problem, research question, hypothesis

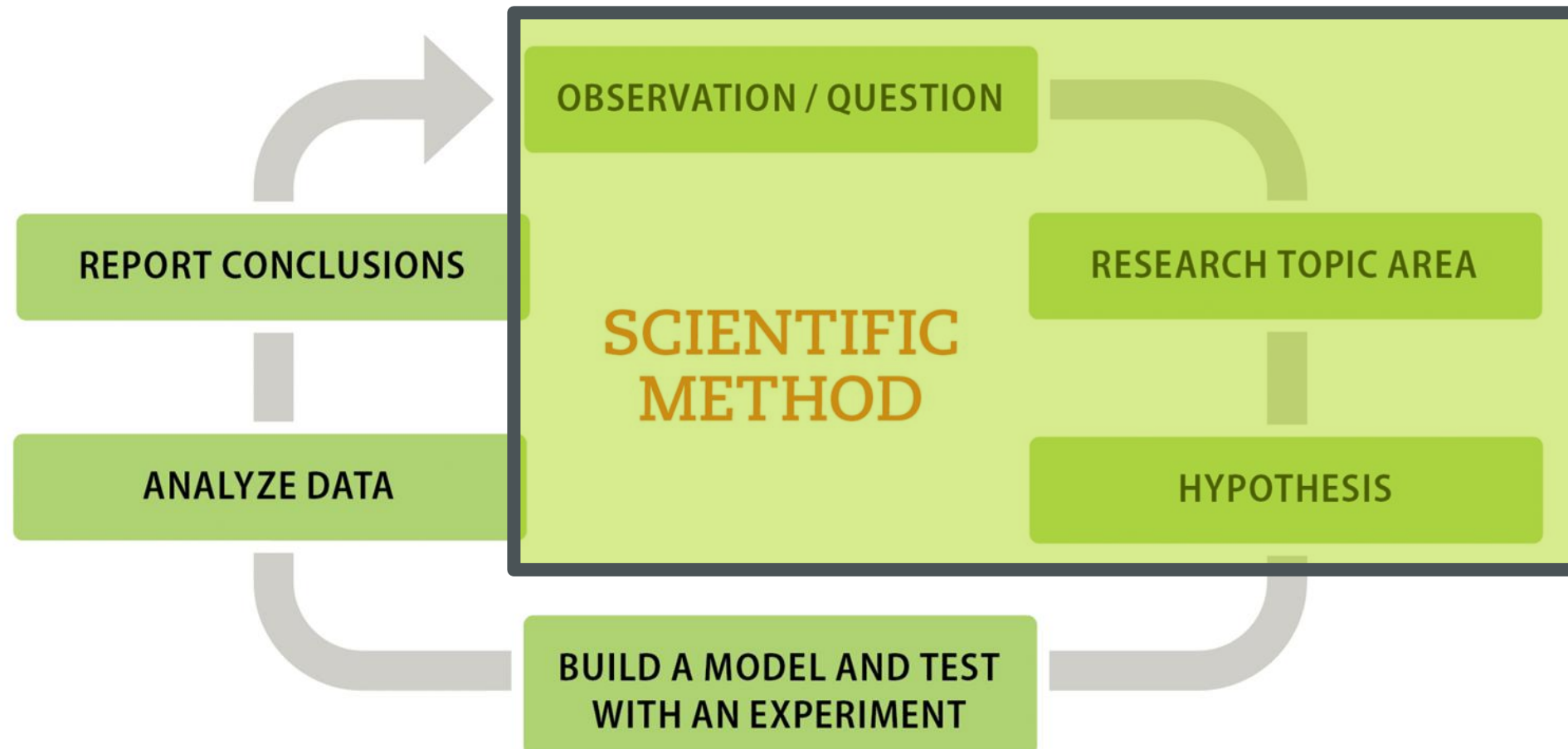
by Alexandre Valle and Daniel Mendes

rev 20260921 draft

- Highlights on the scientific method



- Highlights on the scientific method



● Summary

1. Observations
2. Contextual information
3. Problem
4. Research question
5. Hypothesis
6. Experimentation
7. Result interpretation
8. Conclusion
9. Replication
10. Communication

● Important note to readers

- I. In the following slides, several dissertation excerpts from M.EIC are presented for illustrative purposes. The examples are:

example 1

example 1

example 1

- I. The reader should not consider the provided examples as the single option, nor even the best approach possible. By no means do the examples cover the possible options that depend on the area/topic.
- I. Your supervisor may provide you with other alternative options/approaches.
- I. If you want, you can ask your supervisor to provide you with examples of previous best practices.

● Some definitions

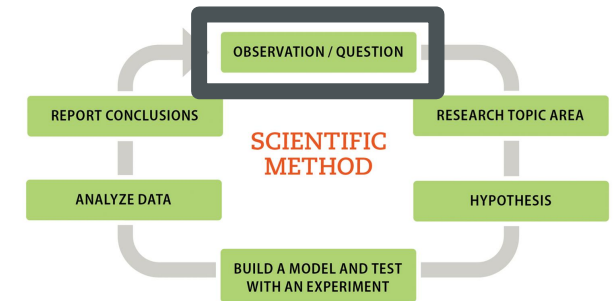
Phenomenon - any observable event or occurrence that can be studied and analyzed (gravity, electromagnetism, thermodynamics, algorithm performance, data structure efficiency, concurrency/parallelism, user interaction)

Tool: any instrument, equipment, methodology, technology used to systematically collect data, conduct experiments, analyze information (profilers, network analyzers, log analyzers, debuggers, performance monitoring, code analysis, simulation, visualization)

Instrument: a specific device designed to observe, measure, and record data.

● Observation

- The initial step of the scientific method. All versions of the scientific method start with a question based on observations.
- Process of systematically noticing and describing events/processes in an organized and consistent manner about a particular phenomenon
- Requires to:
 - a. Identify subject - what to observe
 - b. Collect data - through senses or tools in an organized and consistent manner
 - c. Repeat process - multiple observations improve consistency and reliability
 - d. Objectivity (without biases) - analyze and interpret data



● Observation

- **Empirical observation**

- Based on what is experienced rather than on theory
- Can be verified by experimentation/replication

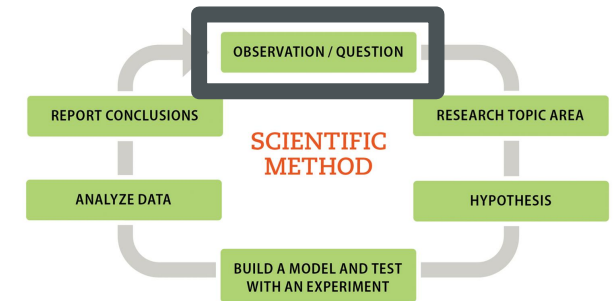
“This theory needs to be backed up with solid empirical data/evidence.”

- **Non-empirical observation**

- Based on personal beliefs, opinions or subjective interpretation rather than on experience
- Difficult to replicate or verify through direct observation.

“This argument is untestable and therefore unempirical.”

“The original experience becomes devalued as subjective and unempirical.”



Serve different purposes and are used in different contexts in scientific research

● Observation

- **Empirical observations:**

- **Quantitative** (using numerical measurements)

- Objective, precise, replicable

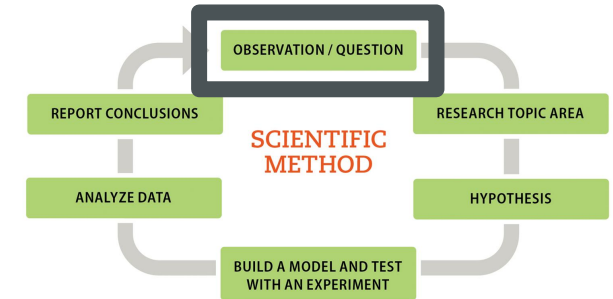
- User interaction, network analysis, system performance, web analytics

- **Qualitative** (without using numerical measurements)

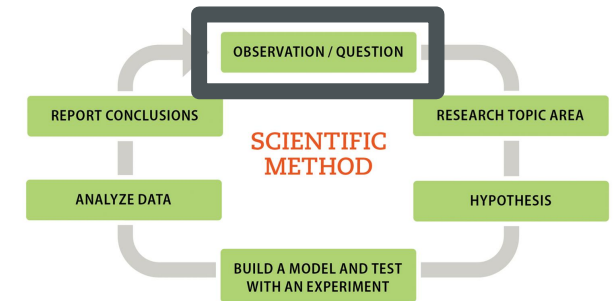
- Descriptive (highly detailed), subjective, contextual

- human/animal behavior, cultural research, botany description

- Rely on direct sensory analysis

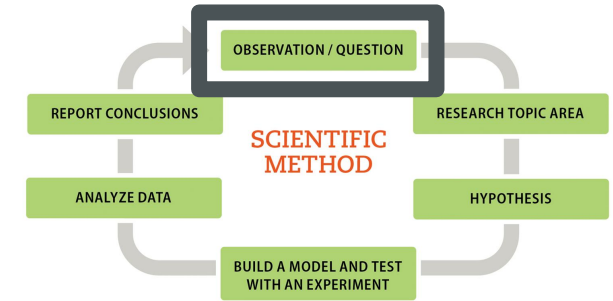


● Background / contextual information



- **Preliminary study** on what is already known
- Communicate other authors' observations/findings
- Cite all the works you used to formulate your ideas. It will ground the problem identification in the existing literature.
- It helps understand the background and contextualise gaps in the existing knowledge.

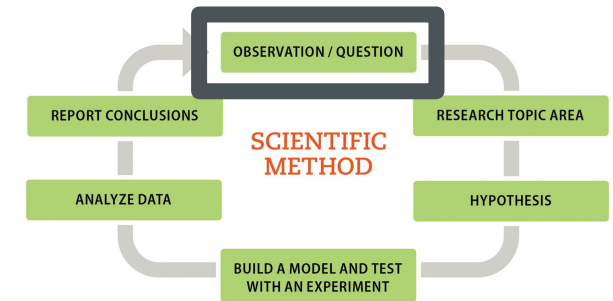
● Background / contextual information



- Some sources (among many other, general, and domain-specific): IEEE Explore, ACM Digital Library, Clarivate, Scopus. Ask your supervisor!
- Some sources require an institutional login. Use your UP credentials!
- Avoid grey literature, for instance,
 - abstracts presented at conferences
 - unpublished/working papers
 - government documents
 - manufacturer information...

● Background / contextual information

- Let you know if others:
 - have done the same experiment
 - what results have they accomplished
 - if there are widely accepted conclusions
 - if there is room for your research:
 - build on top of other's work
 - try a different perspective
 - integrated distinct perspectives
- From a communication perspective, introduces the reader to the **broader context**.



● Background / contextual information (examples)

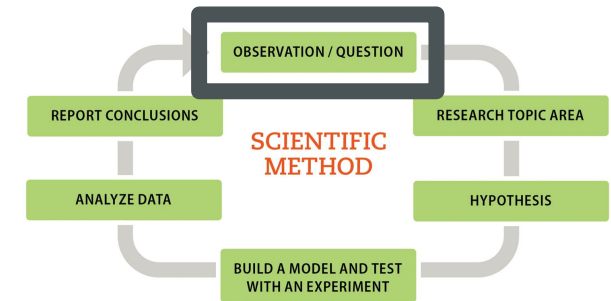
example 1

The exponential growth of information emphasizes the necessity to present it in a user-friendly format, ensuring clarity and comprehension. Many companies share this concern, prompting the question of how to present information to their users in a more appealing and effective way. This task is a daily challenge facing ZOS, Lda., a company that manages a substantial volume of intricate and historical data that must be presented in an attractive and understandable manner. Numerous innovative techniques have been developed globally to convey data through diverse visual formats, making selecting the most fitting approach based on the context and specific data characteristics imperative.

1.1 Context and Motivation

ZOS, Lda. (ZOS) has been informing sports fans and professionals since 2003 [68]. The company has an extensive data collection spanning various sports over the years. This repository is constantly updated and enriched, ensuring its accuracy and relevance. In the realm of soccer, ZOS is the home of the world's largest soccer database [68], featuring comprehensive information on the history of soccer, national teams, major league clubs, and players, as well as those in minor or regional leagues. This wealth of detailed information is maintained through ongoing collaboration with numerous institutions and the contributions of enthusiastic supporters, establishing the company as a leading and trusted authority in the industry.

Considering all available information, it is crucial to present and communicate the multidimensional and multivariate data efficiently. The company maintains a dedicated focus on employing innovative strategies to properly convey information to its users. One effective visual tool used in the sports domain is infographics, which presents complex data in an attractive and easily understandable manner, allowing viewers to identify and analyze the content shown. ZOS aims to present its substantial volumes of information, mainly historical and highly interconnected data, in a more comprehensive and dynamic format capable of attracting and engaging its users.



Background / contextual information (examples)

example 2

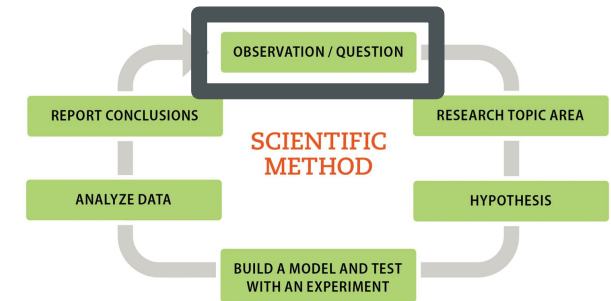
1.1 Context and Motivation

In order to faithfully represent the physicality of a complete virtual environment for the user to interact with, one would have to recreate the whole scene in the real world, which is impractical and less than ideal. Haptic devices overcome this by providing ways to partially represent aspects of the environment and enriching the experience.

Unfortunately, the use of haptics in VR is largely unexplored in commercial solutions, with the most common being wireless controllers or hand trackers, which use buttons, pads, and vibration feedback for input and output. These solutions miss the advantages that having physical devices that more closely reflect the virtual objects might bring, like allowing the user to directly grasp them and perceive them with the sense of touch.

The research works that do explore physical, graspable devices are commonly based on wearable or grounded devices adapted to specific scenarios and situations and thus do not provide an adaptive, unrestraining haptic device. The representation of different objects with the same device is also a limitation often found in haptic solutions due to the difficulty of creating mechanisms and materials for a versatile shape-changing device, able to render objects with various shapes or characteristics [23].

Virtual reality applications (like entertainment, rehabilitation, or training applications, to give some examples) would benefit heavily from having a global and adaptive haptic device because they would be able to immerse the user more effectively, so it is a field that is very worth exploring.



Background / contextual information (examples)

example 3

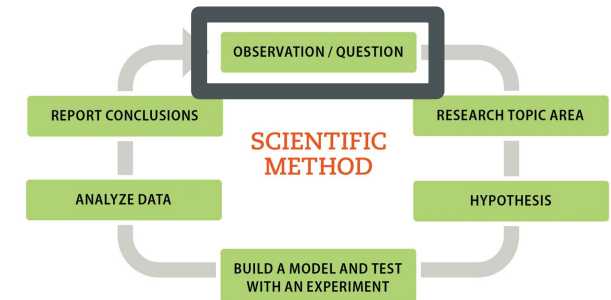
1.1 Context and Motivation

Requirements Engineering (RE) is a vital discipline within software development, encompassing activities such as elicitation, analysis, negotiation, documentation, and validation. However, traditional RE methods often assume direct access to end-users, which is not always true in today's market-driven product development landscape. Furthermore, the cost of conventional RE techniques can be prohibitive, particularly for small startups. A study by Ali and Hong [3] highlighted such companies' budget constraints, underscoring the need for cost-effective alternatives.

Enter social media, a vast and dynamic realm with great promise for RE. According to Statista (2023) [64], 59.4% of the global population uses social media, a significant figure when compared to 64.4% with internet access. Social media platforms host a wealth of information, including users' opinions, experiences with products and services, and expressions of desire for changes in those products and services. This user feedback can be a valuable resource for RE engineers, enabling them to gather requirements and learn more about users through observation.

Moreover, the richness of social media interactions extends beyond individual expressions. Users frequently engage with each other, sharing insights, recommendations, and critiques. This dynamic social environment provides an added layer of information for RE. By analyzing individual posts and user interactions, RE engineers gain a deeper understanding of collective sentiments, preferences, and emerging trends.

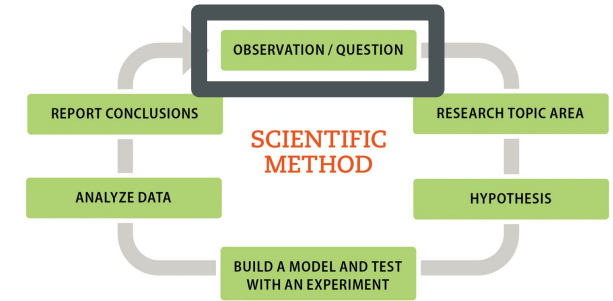
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Reddit and Facebook [28, 47, 51, 74]. The findings of these studies have been conclusive: the gathered feedback not only provides profound insights into user needs but also contributes to the enhancement of software quality. It encompasses invaluable information, spanning bug reports, feature requests, and more [28].

While the extraction of vital feedback began primarily through manual means, the sheer volume of data in social media platforms rendered this approach unsustainable. Consequently, researchers have proposed automatic or semi-automatic mechanisms for classifying and prioritizing feedback. However, these research endeavors have predominantly remained confined to academic purposes and have yet to find practical applications within real-world companies. As a result, a significant gap exists in the current literature: the absence of tools that enable stakeholders to visualize the amassed feedback and make informed decisions based on this wealth of information. It is precisely this gap in the existing literature that this dissertation seeks to explore and address.

● Problem definition



- **Problem statement**

- Using observations, identify a problem
- clearly identify the issue or gap at the centre of your research

- **Justification**

- Explain why this is a problem worth researching
- For whom is the problem relevant
- What would be the advantages of a possible approach

● Problem definition (examples)

example 1

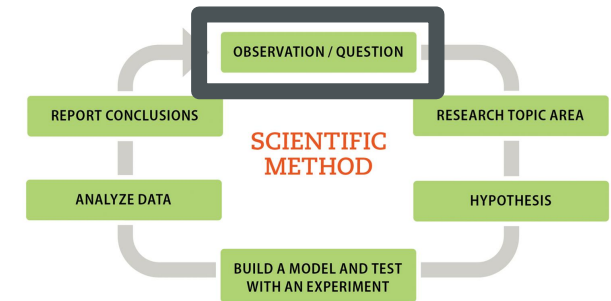
1.2 Problem Identification

The vast and complex realm of sports data presents a unique challenge in making informative, clear, and comprehensive visualizations. The different user profiles and visual literacy levels in the online world further exacerbate this challenge. The complexity lies in sports data's intricate and dynamic nature and its usage in several media outlets. This work focuses on visualizing the temporal evolution of players in national teams and clubs. This problem involves the following dimensions:

- The players may or may not belong to a national team at a specific time. Therefore, one must consider both the players who have affiliations with both national teams and clubs simultaneously, as well as those who exclusively participate at the club level;
- The relationships between players and clubs can change over time, whether through transfers, loans, or retirements.

Capturing and presenting inherently dynamic data as a player, characterized by complex time-evolving relationships, is still not trivial. The challenge is not only to accurately represent this complexity but also to maintain visual consistency throughout the storytelling process [21], providing valuable insights to analysts, coaches, and sports enthusiasts about player transitions, career trajectories, and team dynamics.

Highly connected data refers to complex networks with numerous interdependencies [6] while time-evolving data pertains to data that changes over time [28]. In the context of highly connected and time-evolving information visualization is essential to ensure that the visual representation remains comprehensible and coherent without overwhelming the user. Balancing the need for detail with the imperative of clarity poses a significant challenge that forms the core of the problem addressed in this thesis.

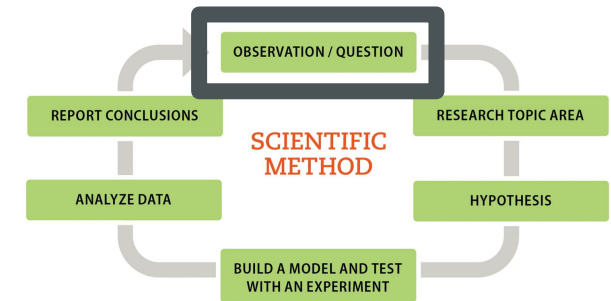


● Problem definition (examples)

example 2

3.1 Problem statement and hypothesis

Upon conducting a comprehensive analysis of the current state-of-the-art and available solutions, we have identified specific areas that present opportunities for further development. Graspable devices provide a more natural way of interacting with devices, as users can pick them up and put them down, as opposed to wearable devices. However, these devices must be manually relocated to the virtual object's target position. We recognise the powerful interactions a graspable device can provide, but we acknowledge their limitations in terms of mobility, which constrains the breadth of possible interactions, mainly dynamic ones. In a pursuit to advance research in this domain, our primary objective is to create a solution that can add the dimension of mobility to existing graspable devices. Considering the identified challenge, the hypothesis that we will be trying to



Problem definition (examples)

example 3

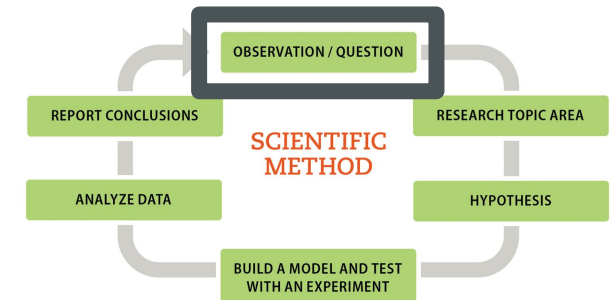
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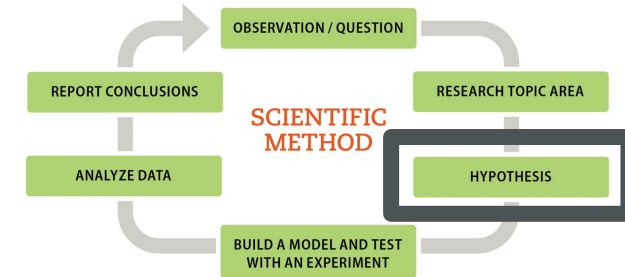
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● Research question



Based on the background research and problem definition, formulate one or more research questions that will guide your research.

- **Unbiased:** don't use personal beliefs to incline or be against some end
- **Clear:** avoid being vague or fuzzy
- **Testable:** you need to achieve evidence
- **Feasible:** you have limited time and resources
- **Ethical:** acceptable according to the rules of a code of conduct

Research questions are open-ended and define the aim to explore.

Research question (examples)

example 1

1.4 Research Questions

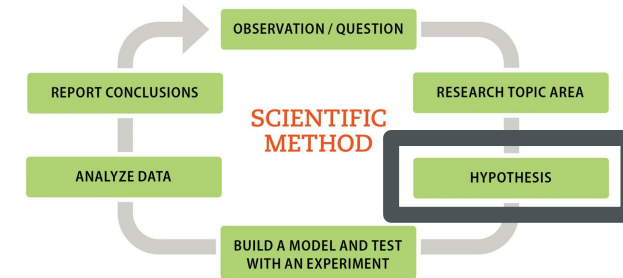
Given the hypothesis outlined in the preceding section, it is imperative to identify and understand visualization techniques and tools that align with the specific characteristics of the data under consideration (highly interconnected data with temporal evolution). It involves exploring graph-based approaches, encompassing their capabilities, limitations, and applications. These objectives can be translated into the following set of research questions:

Q1: Which visualization techniques are commonly used to present highly connected data?

Q2: Which visualization techniques are commonly used to present temporal data?

Q3: Which visualization techniques are commonly used to present multidimensional and multi-variate data?

Q4: What are the limitations and applications of graph-based visualizations for time-evolving data?

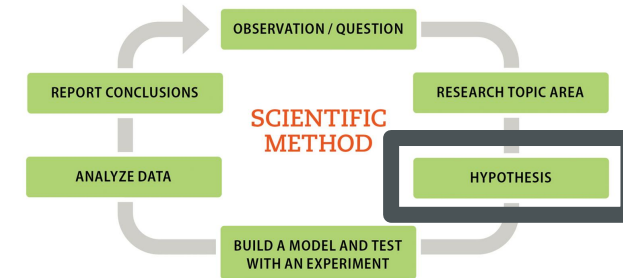


Research question (examples)

example 2

To more clearly understand how we can evaluate our hypothesis, we define the research questions as the following:

- RQ1** Is it possible to simulate a diverse range of virtual objects using a reduced number of mobile devices in a way that remains imperceptible to the user?
- RQ2** What are the limitations on the capabilities of the mobile device that allow it to still precisely replicate the positioning of virtual objects in a timely manner?



Research question (examples)

example 3

Therefore, the goal of this work is to create a solution for the obfuscation of *Wasm* code, employing a variety of obfuscation techniques. To achieve this goal, a comprehensive exploration is necessary to identify techniques compatible with the *Wasm* language. The feasibility of applying these techniques in *Wasm* hinges on ensuring that their implementation maintains the original code's behaviour.

Furthermore, this work seeks to assess the potency and resilience of the obfuscated code after the application of individual techniques or their combinations. Finally, a thorough examination of performance implications is crucial, taking into account factors such as the execution time and code size of the resulting code, as well as evaluating the impact on the code generation process itself, encompassing the time required for its generation.

Hence, to address the problem exposed in Section 1.2, this work aims to address the following questions, regarding the developed obfuscator, called *WasmX* (see Chapter 5):

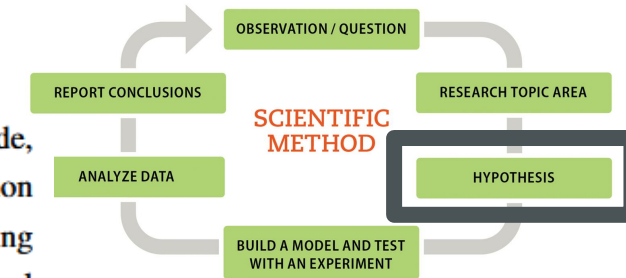
RQ1 Are the programs obfuscated by *WasmX* syntactically correct?

RQ2 Are the programs obfuscated by *WasmX* semantically correct?

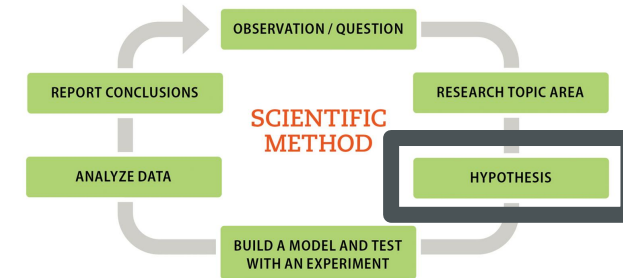
RQ3 Is *WasmX* a time-efficient tool?

RQ4 What is the overhead of *WasmX*'s obfuscation regarding code size and runtime?

RQ5 How strong is the obfuscation performed by *WasmX*, and which techniques are more effective?



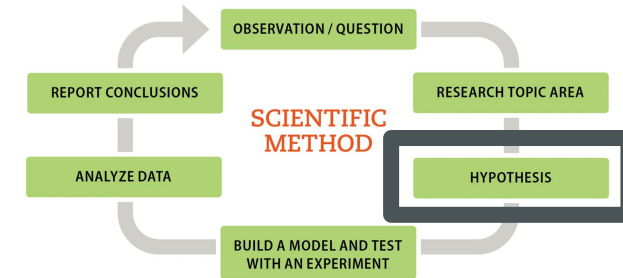
Hypothesis



- **A law** is an explanation in which we have the greatest confidence, based on a long track record of confirmations.
- **A theory**, for most scientists, denotes an explanation that has been confirmed sufficiently to be generally accepted, but which is less firmly established than a law.
- **An axiom** is a concept that is accepted without proof, perhaps because it is obvious or universally accepted (e.g., time, causality) or perhaps to investigate its implications.
- **A hypothesis** is an idea that is still in the process of active testing; it may or may not be correct.
- **A model** is a mathematical or conceptual hypothesis that provide useful perspectives despite recognized oversimplification.

Hypothesis

- Once you have a clear research question, you may formulate hypothesis (optional for master dissertations).
- A testable statement that predicts a possible answer to your research question.
- It predicts the outcome of the experiment based on your knowledge and research performed so far.
- In the end:
 - **Confirmation of the hypothesis merely demonstrates it is consistent with observations**, thereby increasing confidence that the hypothesis is correct.
 - **Hypotheses have not fully bridged the gap between the particular and the universal**: they are tied to the observations that they try to explain.



● Hypothesis: is it required or not?

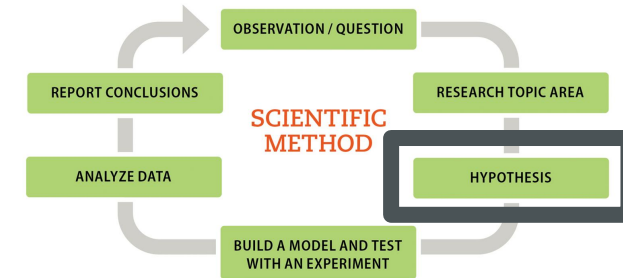
Some supervisor will not require hypothesis definition. It may be considered optional for a master (but mandatory for a PhD).

Research that welcomes a hypothesis:

- Quantitative research

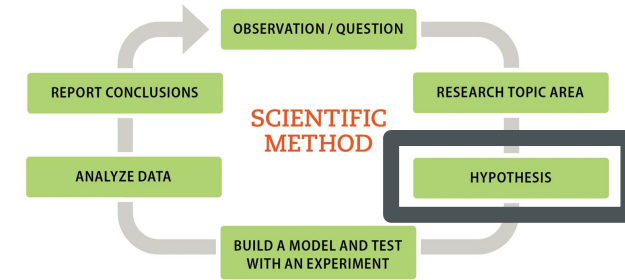
Research that may not require a hypothesis (research question(s) suffice):

- Qualitative or exploratory research
- Descriptive studies /descriptive analysis (literature, history)

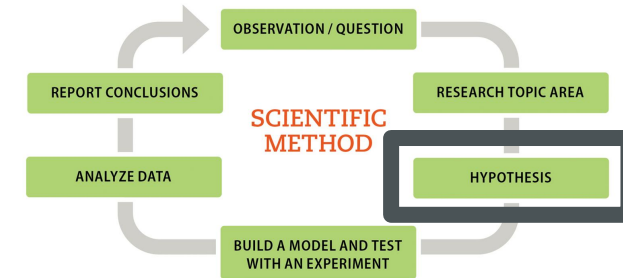


Hypothesis

- Relevant and original
- Unbiased
- Specific and simple
- Testable
- Feasible
- Ethical



Hypothesis



- in the end, a Hypothesis can be:
 - accepted: if the results support the hypothesis
 - rejected: if the results do not support the hypothesis
- you can have a hypothesis rejected and still having applied the entire process correctly

Note: to change the hypothesis if it proves rejected defeats the whole purpose of the scientific method.

Hypothesis (optional, as said)

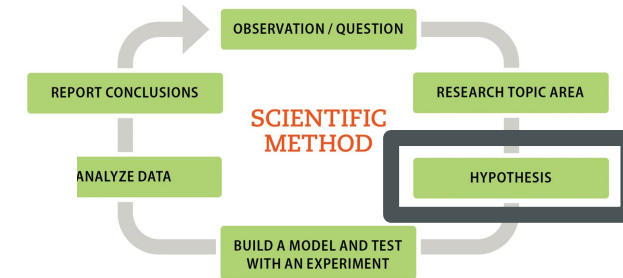
example 1

1.3 Hypothesis

Considering the problem previously identified, this work formulates the following hypothesis:

Graph-based visualizations for time-evolving data are well suited for presenting highly connected entities with complex relationships while preserving the story within the visualization.

Graphs are renowned for their capacity to capture complex relationships and temporal dependencies within data [28]. The historical information and dynamic changes over time can be represented using animations [11]. This results in a visualization method tailored to the intricate and dynamic nature of complex, hyper-connected data enriched with an additional temporal dimension.



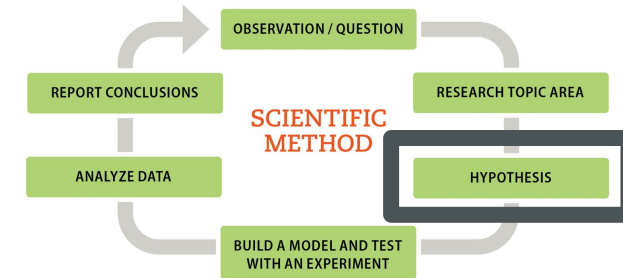
Hypothesis (optional, as said)

example 2

Upon conducting a comprehensive analysis of the current state-of-the-art and available solutions, we have identified specific areas that present opportunities for further development. Graspable devices provide a more natural way of interacting with devices, as users can pick them up and put them down, as opposed to wearable devices. However, these devices must be manually relocated to the virtual object's target position. We recognise the powerful interactions a graspable device can provide, but we acknowledge their limitations in terms of mobility, which constrains the breadth of possible interactions, mainly dynamic ones. In a pursuit to advance research in this domain, our primary objective is to create a solution that can add the dimension of mobility to existing graspable devices. Considering the identified challenge, the hypothesis that we will be trying to test and evaluate is defined as follows:

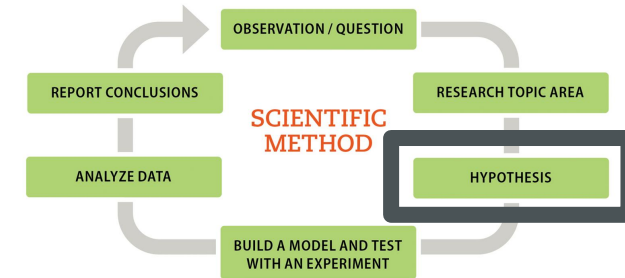
It is possible to simulate multiple virtual objects in a convincing way using a smaller number of movable devices.

Several challenges must be overcome to address the hypothesis. Two primary issues stand out in the idealisation of a device that targets this hypothesis. The first challenge involves accurately predicting which object the user intends to interact with so that the devices understand which virtual object to replicate. The second challenge lies in accurately tracking the device's position and moving it to the target position, replicating the virtual object's location. Both of these issues pose substantial challenges, but we believe in prioritising the resolution of the challenge of accurate positioning and movement, given its central role in our device's functionality. To navigate this



so... what comes next?

What next?



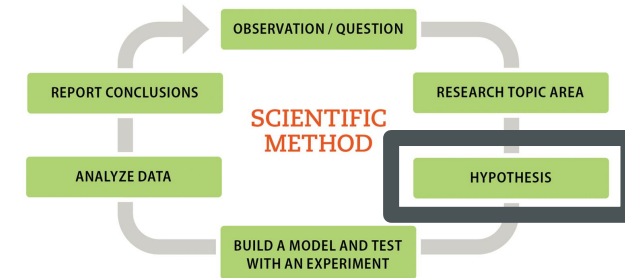
For the thesis manifest, you are required to have:

1. Context/motivation
2. Problem definition
3. Research questions (one or more)
4. Hypothesis (optional)

NOTES:

- a) Some supervisors may suggest other names for sections
- b) 3 and 4 can be replaced by Objectives or Research Questions (but research questions are highly advisable)

What next?



Later, you will need to conduct a literature review to:

- improve your knowledge of the domain and topic
- know what other authors did before
- identify gaps, limitations, and weaknesses.

After, you will have a clearer idea of what your contribution can be, and you can do/improve context/problem / RQ / (hypothesis) / (other sections).

These contents are the vision of your work!

Everything in your work beyond this stage is a consequence of this vision.

What next?

Table 1 Main review types characterized by methods used

Label	Description	Methods used (SALSA)			
		Search	Appraisal	Synthesis	Analysis
Critical review	Aims to demonstrate writer has extensively researched literature and critically evaluated its quality. Goes beyond mere description to include degree of analysis and conceptual innovation. Typically results in hypothesis or model	Seeks to identify most significant items in the field	No formal quality assessment. Attempts to evaluate according to contribution	Typically narrative, perhaps conceptual or chronological	Significant component: seeks to identify conceptual contribution to embody existing or derive new theory
Literature review	Generic term: published materials that provide examination of recent or current literature. Can cover wide range of subjects at various levels of completeness and comprehensiveness. May include research findings	May or may not include comprehensive searching	May or may not include quality assessment	Typically narrative	Analysis may be chronological, conceptual, thematic, etc.
Mapping review/ systematic map	Map out and categorize existing literature from which to commission further reviews and/or primary research by identifying gaps in research literature	Completeness of searching determined by time/scope constraints	No formal quality assessment	May be graphical and tabular	Characterizes quantity and quality of literature, perhaps by study design and other key features. May identify need for primary or secondary research
Meta-analysis	Technique that statistically combines the results of quantitative studies to provide a more precise effect of the results	Aims for exhaustive, comprehensive searching. May use funnel plot to assess completeness	Quality assessment may determine inclusion/exclusion and/or sensitivity analyses	Graphical and tabular with narrative commentary	Numerical analysis of measures of effect assuming absence of heterogeneity
Mixed studies review/mixed methods review	Refers to any combination of methods where one significant component is a literature review (usually systematic). Within a review context it refers to a combination of review approaches for example combining quantitative with qualitative research or outcome with process studies	Requires either very sensitive search to retrieve all studies or separately conceived quantitative and qualitative strategies	Requires either a generic appraisal instrument or separate appraisal processes with corresponding checklists	Typically both components will be presented as narrative and in tables. May also employ graphical means of integrating quantitative and qualitative studies	Analysis may characterise both literatures and look for correlations between characteristics or use gap analysis to identify aspects absent in one literature but missing in the other
Overview	Generic term: summary of the [medical] literature that attempts to survey the literature and describe its characteristics	May or may not include comprehensive searching (depends whether systematic overview or not)	May or may not include quality assessment (depends whether systematic overview or not)	Synthesis depends on whether systematic or not. Typically narrative but may include tabular features	Analysis may be chronological, conceptual, thematic, etc.
Qualitative systematic review/qualitative evidence synthesis	Method for integrating or comparing the findings from qualitative studies. It looks for 'themes' or 'constructs' that lie in or across individual qualitative studies	May employ selective or purposive sampling	Quality assessment typically used to mediate messages not for inclusion/exclusion	Qualitative, narrative synthesis	Thematic analysis, may include conceptual models

A typology of reviews: an analysis of 14 review types and associated methodologies, Maria J. Grant* & Andrew Booth†, *Salford Centre for Nursing, Midwifery and Collaborative Research (SCNMCR), University of Salford, Salford, UK, †School of Health and Related Research (ScHARR), University of Sheffield, Sheffield, UK

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What next?

Table 1 Continued

Label	Description	Methods used (SALSA)			
		Search	Appraisal	Synthesis	Analysis
Rapid review	Assessment of what is already known about a policy or practice issue, by using systematic review methods to search and critically appraise existing research	Completeness of searching determined by time constraints	Time-limited formal quality assessment	Typically narrative and tabular	Quantities of literature and overall quality/direction of effect of literature
Scoping review	Preliminary assessment of potential size and scope of available research literature. Aims to identify nature and extent of research evidence (usually including ongoing research)	Completeness of searching determined by time/scope constraints. May include research in progress	No formal quality assessment	Typically tabular with some narrative commentary	Characterizes quantity and quality of literature, perhaps by study design and other key features. Attempts to specify a viable review
State-of-the-art review	Tend to address more current matters in contrast to other combined retrospective and current approaches. May offer new perspectives on issue or point out area for further research	Aims for comprehensive searching of current literature	No formal quality assessment	Typically narrative, may have tabular accompaniment	Current state of knowledge and priorities for future investigation and research
Systematic review	Seeks to systematically search for, appraise and synthesis research evidence, often adhering to guidelines on the conduct of a review	Aims for exhaustive, comprehensive searching	Quality assessment may determine inclusion/exclusion	Typically narrative with tabular accompaniment	What is known; recommendations for practice. What remains unknown; uncertainty around findings, recommendations for future research
Systematic search and review	Combines strengths of critical review with a comprehensive search process. Typically addresses broad questions to produce 'best evidence synthesis'	Aims for exhaustive, comprehensive searching	May or may not include quality assessment	Minimal narrative, tabular summary of studies	What is known; recommendations for practice. Limitations
Systematized review	Attempt to include elements of systematic review process while stopping short of systematic review. Typically conducted as postgraduate student assignment	May or may not include comprehensive searching	May or may not include quality assessment	Typically narrative with tabular accompaniment	What is known; uncertainty around findings; limitations of methodology
Umbrella review	Specifically refers to review compiling evidence from multiple reviews into one accessible and usable document. Focuses on broad condition or problem for which there are competing interventions and highlights reviews that address these interventions and their results	Identification of component reviews, but no search for primary studies	Quality assessment of studies within component reviews and/or of reviews themselves	Graphical and tabular with narrative commentary	What is known; recommendations for practice. What remains unknown; recommendations for future research

What next?

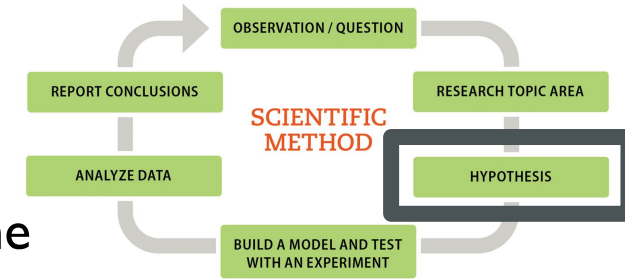
Context / problem / RQ / (hypothesis) / (other sections) will compose the Chapter I - Introduction, of your thesis.

Later, for Chapter I you can also add:

5. **Methods / Methodology**¹ (write it when you have a clear view of your experiment)
6. **Contributions** (write it after the work done, you can state the contributions)
7. **Document structure** (add it at the end of the dissertation document writing)

¹Method: a way of doing anything, especially according to a defined and regular plan

Methodology: the branch of knowledge that deals with methods generally or with the methods in a particular discipline of field of study. Source: [Oxford English Dictionary](#)



Talk to your supervisor!

Dissertation Preparation

• Questions?